

FPGA Programming – VHDL Part

VHDL by Examples – Hello World

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Hardware description

- Drawing a diagram of the hardware design (schematics)
- Textual description
 - Programming language that include explicitly the notion of time
 - Concurrent language
 - Implement the Register Transfer Level (RTL) of a circuit (not dependent on hardware technology but only dataflow between registers and logical operations)
 - Most widely used
 - Verilog
 - VHDL [VHSIC (Very High Speed Integrated Circuit) Hardware Description Language]

VHDL basic structures

- **entity**: it is a black box where only the interface signals (**ports**) are described
- **architecture**: it describes the content of the box in terms of functionalities and/or structures of the circuits

```
entity And2 is  
    port (x,y: in BIT; z: out BIT);  
end entity And2;
```

```
architecture ex1 of And2 is  
begin  
    z <= x and y;  
end architecture ex1;
```

```
architecture ex2 of And2 is  
begin  
    z <= '1' when x & y = "11" else  
        '0';  
end architecture ex2;
```

Standard logic

'U' Uninitialized

'X' Forcing (i.e. strong) unknown

'0' Forcing 0

'1' Forcing 1

'Z' High impedance

'W' Weak unknown

'L' Weak 0

'H' Weak 1

'-' Don't care

	U	X	0	1	Z	W	L	H	-
U	U	U	U	U	U	U	U	U	U
X	U	X	X	X	X	X	X	X	X
0	U	X	0	X	0	0	0	0	X
1	U	X	X	1	1	1	1	1	X
Z	U	X	0	1	Z	W	L	H	X
W	U	X	0	1	W	W	W	W	X
L	U	X	0	1	L	W	L	W	X
H	U	X	0	1	H	W	W	H	X
-	U	X	X	X	X	X	X	X	X

VHDL basic structures

- **library declaration**
- **entity**: it is a black box where only the interface signals (**ports**) are described
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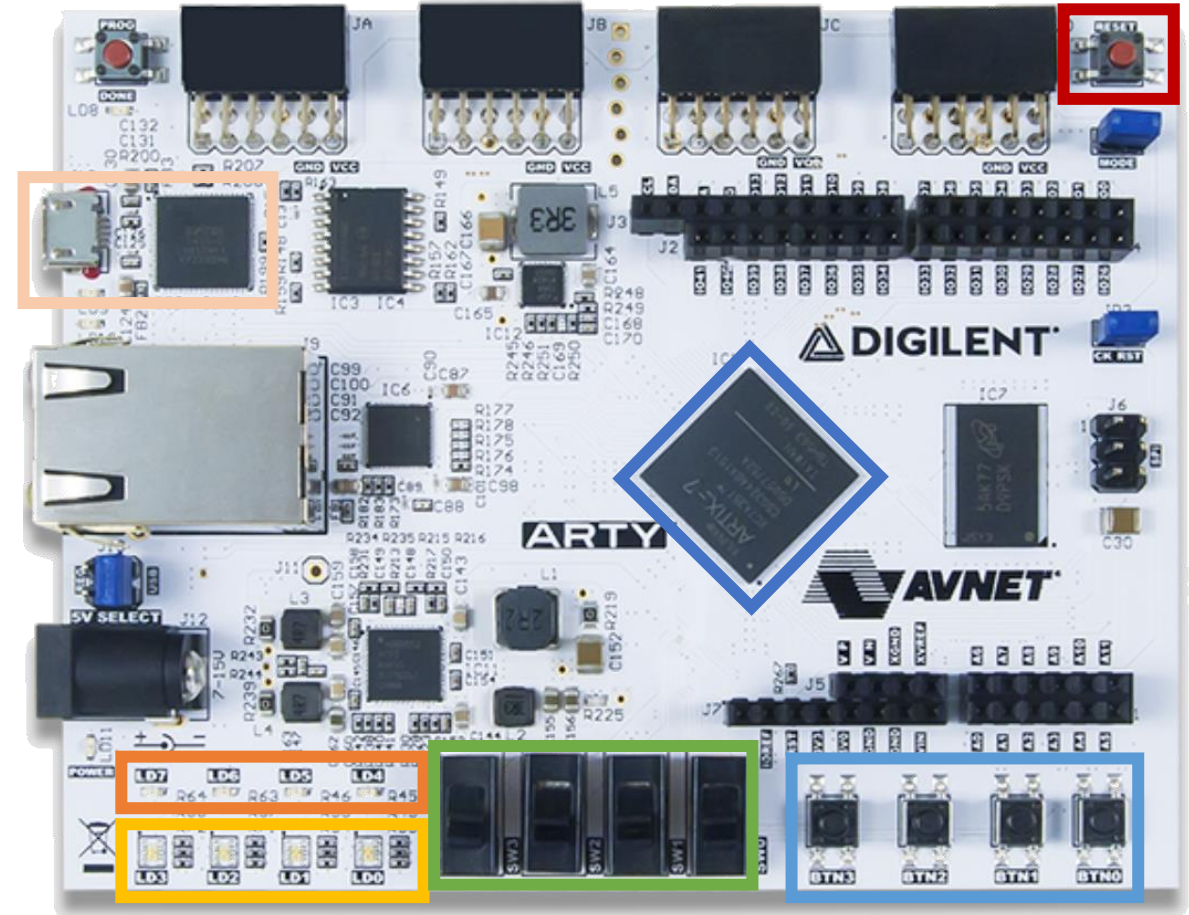
```
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;
```

```
entity And2 is  
    port (x,y: in std_logic;  
          z: out std_logic);  
end entity And2;
```

```
architecture ex1 of And2 is  
begin  
    z <= x and y;  
end architecture ex1;
```

ARTY A7

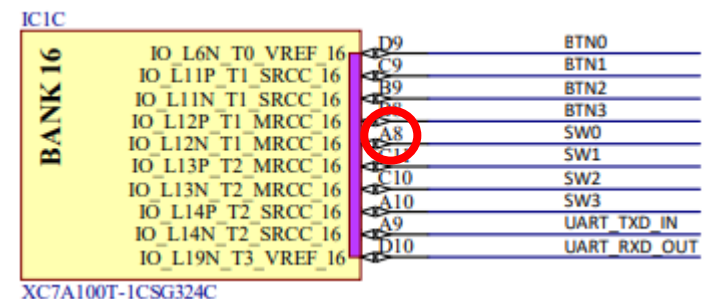
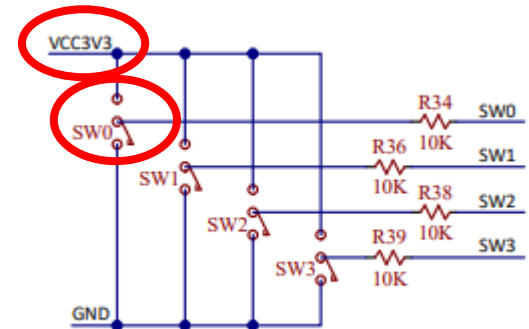
- Artix-7 FPGA
 - [XC7A100TCSG324-1](#)
 - 15,850 slices
 - 4,860 Kbits BRAM
 - 6 CMTs
 - 240 DSP
- ARTY A7 board
 - 1 [USB-UART/JTAG Bridge](#)
 - 4 [Switches](#)
 - 4 [Buttons](#)
 - 1 [Reset Button](#)
 - 4 [LEDs](#)
 - 4 [RGB LEDs](#)
- [Reference manual](#)
- [Constraints file](#)
- [Schematics](#)



Minimal constraint file

- At least the pin assignment is needed
 - The board manufacturer could provide a master constraint file
 - Use the schematic

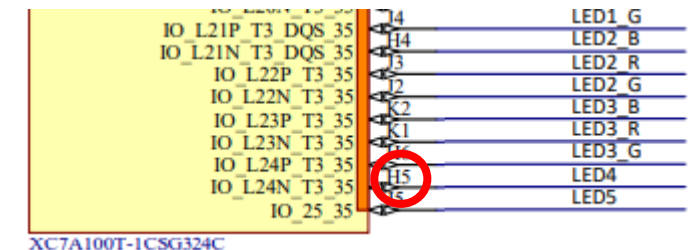
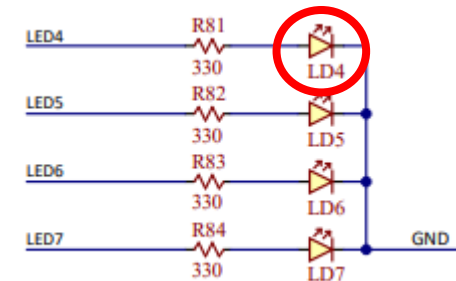
```
set_property PACKAGE_PIN A8 [get_ports { x }];  
set_property IOSTANDARD LVCMOS33 [get_ports { x }];  
  
set_property PACKAGE_PIN C11 [get_ports { y }];  
set_property IOSTANDARD LVCMOS33 [get_ports { y }];  
  
set_property PACKAGE_PIN H5 [get_ports { z }];  
set_property IOSTANDARD LVCMOS33 [get_ports { z }];
```

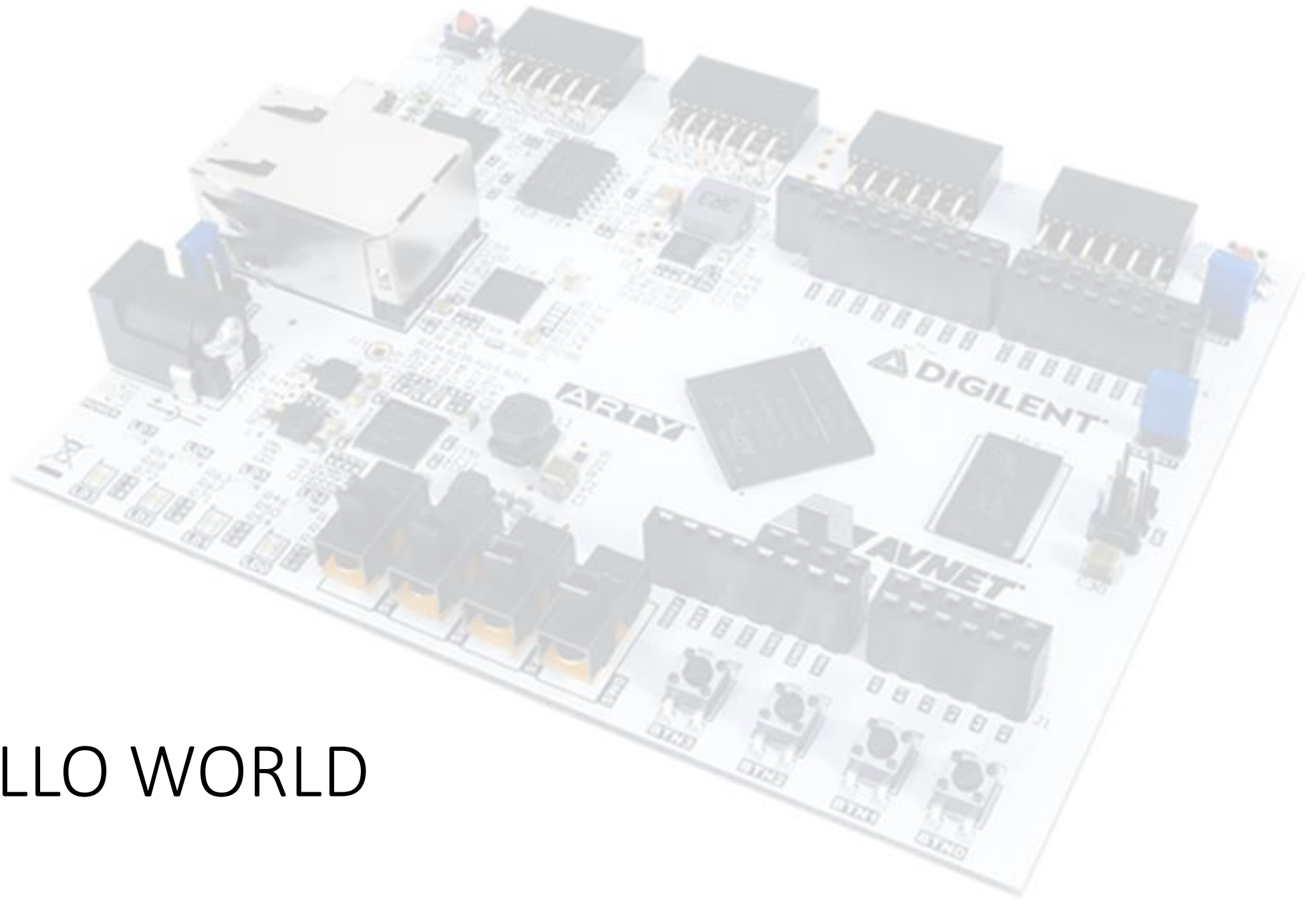


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LAB HELLO WORLD